MATH 1210 Worksheet 6 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. Consider the matrix
$$A = \begin{pmatrix} 2i & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 1 \\ 1 & -i & \sqrt{2} \end{pmatrix}^4.$$
Compute the determinant of A. *Hint: Use that* $|AB| = |A||B|$

2. Use Gaussian elimination to compute the determinant of the matrix
$$A = \begin{pmatrix} 2 & 3 & 2 & -1 & -2 \\ 0 & -1 & -1 & 1 & 0 \\ -3 & 1 & 1 & 2 & 1 \\ 1 & -2 & 2 & 1 & -1 \\ 0 & 1 & 1 & 2 & -1 \end{pmatrix}$$

3. For a real number x consider the matrix
$$A = \begin{pmatrix} 1 & x & 2 \\ x & 0 & 2 \\ 1 & 1 & 1+x \end{pmatrix}.$$
   (a) Compute the determinant of A for $x = 1$.
   (b) Find all values of x for which $|A| = 0$. *Hint: use part (a).*

4. Use Cramer's rule to solve the system
$$\begin{aligned} 2x - 3y + 6z &= 1 \\ 3x + y - z &= 0 \\ x + y - 2z &= -1. \end{aligned}$$

5. Consider the system
$$\begin{aligned} ax_1 + 6x_3 &= 12 \\ x_1 + 3x_2 + 4x_3 &= -1 \\ x_2 + x_3 &= b, \end{aligned}$$
where a and b are two real numbers.
   (a) For which value(s) of a and b does the system have a unique solution?
   (b) For which value(s) of a and b does the system have at least two solutions?

6. Consider the system
$$\begin{aligned} 2x + z &= 3 \\ x + 2y + tz &= 5 \\ -x + 2y &= 4 \end{aligned}$$
where t is a real number. It is given to you that the system has a unique solution (x, y, z) with $z = 1$. Use Cramer's rule to find t.

7. Determine if each of the following set of vectors are linearly independent.
   (a) $\mathbf{u}_1 = \langle 1, -3, 7 \rangle$, $\mathbf{u}_2 = \langle 5, 2, 3 \rangle$, $\mathbf{u}_3 = \langle 13, -7, -1 \rangle$, $\mathbf{u}_4 = \langle 1, 1, -1 \rangle$.
   (b) $\mathbf{u}_1 = \langle 2, 0, 3 \rangle$, $\mathbf{u}_2 = \langle 1, -1, 3 \rangle$, $\mathbf{u}_3 = \langle 6, -2, 2 \rangle$.

8. (a) Are the following three vectors

$$\mathbf{u}_1 = \langle 1, 5, 2, 2 \rangle, \quad \mathbf{u}_2 = \langle 2, 4, 1, 2 \rangle, \quad \mathbf{u}_3 = \langle -1, 1, 0, -1 \rangle$$

   linearly dependent?
   (b) Write the vector $\mathbf{v} = \langle 3, 3, -1, 1 \rangle$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Justify your answer.

9. Let $\mathbf{u}$ and $\mathbf{v}$ be two nonparallel nonzero vectors with three components. Show that the vectors $\mathbf{u}$,$\mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ are linearly independent.

10. Consider the vectors

$$\mathbf{u}_1 = \langle 1, 2, 0, 0 \rangle, \qquad \mathbf{u}_2 = \langle -2, 1, 0, -7 \rangle, \qquad \mathbf{u}_3 = \langle 13, 0, 2, 0 \rangle, \qquad \mathbf{u}_4 = \langle 1, 0, 0, 3 \rangle.$$

   Suppose that the vector $\mathbf{v} = \langle 4, 0, 1, 0 \rangle$ can be written as a linear combination

$$C_1\mathbf{u}_1 + C_2\mathbf{u}_2 + C_3\mathbf{u}_3 + C_4\mathbf{u}_4 = \mathbf{v}.$$

   Use Cramer's rule to find C_2. You are not allowed to use any other method.